MIL-I-81023D(AS)
20 December 1985
SUPERSEDING
MIL-I-81023C(AS)
10 March 1977

MILITARY SPECIFICATION

INDUCTOR, 28 V.D.C. LABORATORY TEST GENERAL SPECIFICATION FOR

This specification is approved by use within Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for iron-core inductors to be used as laboratory test equipment for military switches, relays and circuit breakers with inductive current ratings up to 100 amperes.

1.2 Classification. Iron-core inductors shall be classified into types.

1.2.1 Types. Iron-core inductors shall be of the following types as specified (see 6.2).

Type I - Iron-core inductors for use with military switches, relays and circuit breakers with inductive rating of 2 to 100 amperes.

Type II - Iron-core inductors for use with military switches, relays and circuit breakers with inductive rating of less than 2 amperes.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electrical Components Parts
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4).

3.2 Examination and inspection. The inductors furnished under this specification shall be a product which has successfully passed the examination and inspection tests specified herein.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion under normal atmospheric conditions to which the inductors will be subjected in storage or normal service use.

3.3.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.3.3 Selection of materials. Specifications and standards for all materials, parts, government certification, and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.3.3.1 Standard parts. Standard parts (MS, AN or MIL) shall be used wherever they are suitable for the purpose and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts and cotter pins may be used provided they possess suitable properties and are replaceable by the standard parts (MS, AN or MIL) without alteration, and provided the corresponding standard part numbers are referenced in the parts list, and if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Design and construction.

3.4.1 General. Inductors shall be of the design, construction and physical dimensions specified in Figure 1 or Figure 12 as applicable.

3.4.2 Shims. Shims used for spacing magnetic gap shall be a non-magnetic type of stainless steel, brass or bronze.

3.4.3 Thru bolts. Thru bolts shall be of steel and shall be riveted to provide tamperproof construction.

3.4.4 Mounting surface. The mounting surface shall be flat within .015-inch.

3.4.5 Resistance. When inductors are tested as specified in 4.3.2, the terminal to terminal resistance shall be as follows:

Type I - .277 ohm ±10 percent
Type II - 8.15 ohm ±10 percent

3.4.6 Response curves (current versus time).

Type I Inductors - When inductors are calibrated or tested as specified in 4.3.3 the current versus time curves shall fall within the limits shown on Figures 13 and 14. For information purposes, stored energy curves (calculated from the current versus time curves) are shown on Figure 15.

Type II Inductors - The current versus time response shall be as specified in 4.3.3 when calibrated as specified in 4.3.3.

3.4.7 Temperature rise. The temperature of the coil shall not exceed 85°C rise when tested in accordance with paragraph 4.3.4.

3.4.8 Terminals. Terminals shall be as specified in Figure 1 and Figure 12 for Type I and Type II inductors, respectively.

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3.4.8.1 Strength of threaded terminal inserts. The inductor shall operate electrically after the terminal strength test of 4.3.5. There shall be no short-circuiting, breakage, loosening or rotation of terminals.

3.4.9 Mounting. The mounting dimensions shall be as specified in Figure 1 and Figure 12 for Type I and Type II inductors, respectively.

3.4.10 Dielectric withstanding voltage. When tested as specified in 4.3.6, inductors shall withstand the application of the specified voltage without arcing, flashover, breakdown of insulation or damage.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.6 Identification. Inductors shall be marked in accordance with MIL-STD-130, this specification, and with the manufacturer's name and trademark in any convenient location.

3.7 Dimensions. The dimensions shall be as specified in Figure 1 and Figure 12 for Type I and Type II inductors, respectively.

3.8 Workmanship. Inductors shall be processed in such a manner as to be uniform in quality and shall be free from cracked or displaced parts and other defects that will affect life, serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all tests required by this specification shall be made at room ambient conditions of temperature, pressure and humidity.

4.4 First article inspection. First article inspection shall be performed under conditions acceptable to the Government on sample units produced with equipment and procedures normally used in production. All requirements and inspections of this specification shall be met. One sample shall be required.

4.5 Quality conformance inspection.

4.5.1 Quality conformance inspection. All units shall be subjected to the Quality Conformance Inspection specified in Table I in the order shown. Any inductor failing any inspection or test in the sequence will not be subjected to further inspection.

TABLE I. Quality conformance inspection.

Examination or test	Requirement paragraph	Method paragraph
Visual and mechanical	3.1	4.6.1
Resistance	3.4.5	4.6.2
Response (current versus time)	3.4.6	4.6.3
Temperature rise 1/	3.4.7	4.6.4
Terminal strength 1/	3.4.8.1	4.6.5
Dielectric	3.4.10	4.6.6

1/ Test shall be performed on only two inductors of each inspection lot.

4.5.2 Failure. Failure in any of the applicable examinations or tests will be cause for rejection.

4.5.3 Inspection lot. An inspection lot, as far as practicable, shall consist of all the inductors of the same style, design and construction, produced under essentially the same conditions and offered for inspection at one time.

4.5.4 Sampling plan. One-hundred percent of production shall be tested in accordance with Table I. An inductor failing any test in Table I shall not be delivered on the order.

4.5.6 Noncompliance. If a unit fails to pass Quality Conformance Inspection, the supplier shall take corrective action on the materials or process, or both, as warranted, and on all units of product which can be

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corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action has been taken and accepted by the government. After the corrective action has been taken, inspection shall be repeated on additional units (all acceptance inspections or the inspection which the original sample failed, at the option of the government). In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the government.

4.6 Methods of inspection.

4.6.1 Visual and mechanical. Inductors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.2 Resistance. Inductors shall be exposed to a temperature of 25°C ±2°C for a minimum of 4 hours with coil de-energized. The coil resistance measured with a Wheatstone bridge or other suitable means shall be as specified in 3.4.5.

4.6.3 Response curves (current versus time).

Type I - The inductor current versus time characteristics curve shall be obtained by means of an oscilloscope and suitable photographic means at 5.0 amperes and at 25.0 amperes steady state current. A variable resistance of sufficient capacity shall be connected in series with the inductor, and the steady-state current shall be adjusted to each specified current value with power furnished by a suitable 28 ±.5 volt direct current supply. The current versus time curve shall be obtained upon closing the inductor circuit with a suitable switching device whose contact-bounce duration shall be less than 1 millisecond. The response shall be within the limits shown in Figure 13 for 5.0 ±0.5 amperes steady-state current and within the limits shown in Figure 14 for 25.0 ±0.5 amperes steady-state current. During calibration, if the current response is above the response current (inductance too low), reduce shim thickness. If the current response is below the response current (inductance too large), increase shim thickness. After calibration all thru bolts shall be riveted to provide tamper-proof construction. With 28 ±.5 volts applied, the current shall be set to the specified ampere value by adjusting the series resistance and not by adjusting the voltage.

Type II - With .005 inch thick shim washers between each core and the magnetic upper plate as shown in Figure 2, determine the current versus time characteristics of the inductor at 1 amp 28 vdc using appropriate series resistance for current adjustment and an oscilloscope. The current versus time characteristics shall be recorded by monitoring the voltage drop across the series resistance when energizing this inductive load with a bounce-free, switching device. Adjust the air gap of the inductor by adding successive (.002-inch thick) shim washers between each core and the magnetic

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upper plate until the 95 percent steady-state current point (3L/R) is reached in 135 ±5 msec. The .25-inch screw which secures the upper magnetic plate to the cores must remain tight during and following this calibration.

4.6.4 Temperature rise. The coil temperature rise test shall be conducted while the coil is being energized at the rate of 10 times per minute at a duty cycle of 25 percent "ON" and 75 percent "OFF," 28 ±2 volts d.c. (current to be 25 amperes for Type I inductors, 1.95 amperes for Type II inductors). The temperature shall be determined by comparing the resistance of the winding at a known temperature (room ambient) with the resistance of the coil after its temperature has stabilized on the above duty cycle. The temperature of the coil shall be considered stabilized after three identical resistance readings taken at one-half hour intervals are obtained. This shall constitute the maximum resistance value for basis of calculating final temperature using the following formula:

$$T = \frac{R}{r}(234.5 + t) - 234.5$$

where: T = known temperature in degrees C
r = resistance at the known temperature
R = observed resistance
T = temperature sought

The temperature rise is the difference between the calculated temperature and the known temperature. The rise shall not exceed 85°C.

4.6.5 Terminal strength.

a. Threaded terminals shall withstand a static force of 50 lbs for type 1 inductors, and 25 lbs for type 2 inductors. The force shall be gradually applied in any direction.

b. Threaded terminals shall withstand a torque of 75 lb-in. for type 1 inductors and 5 lb-in. for type 2 inductors applied to the head of the terminal screw in a plane parallel to the face of the terminal.

4.6.6 Dielectric withstanding voltage. Inductors shall be tested in accordance with Method 301 of MIL-STD-202. The following exception shall apply:

a. Magnitude and source of test voltage: 1000 volts RMS at 60 cycles a.c.

Upon completion of the test, the inductors shall meet the requirements of paragraph 3.4.10.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Inductors shall be afforded preservation and packaging in accordance with the manufacturer's practice of shipment on a commercial basis.

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5.2 Packing. Inductors packaged in accordance with 5.1 shall be packed in containers of the type, size and kind commonly used for the purpose.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 The inductors covered by this specification are intended for use in laboratory testing of military switches, relays and circuit breakers with d.c. inductive loads.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Type.

c. Packaging and packing instructions if other than as specified (see 5.1).

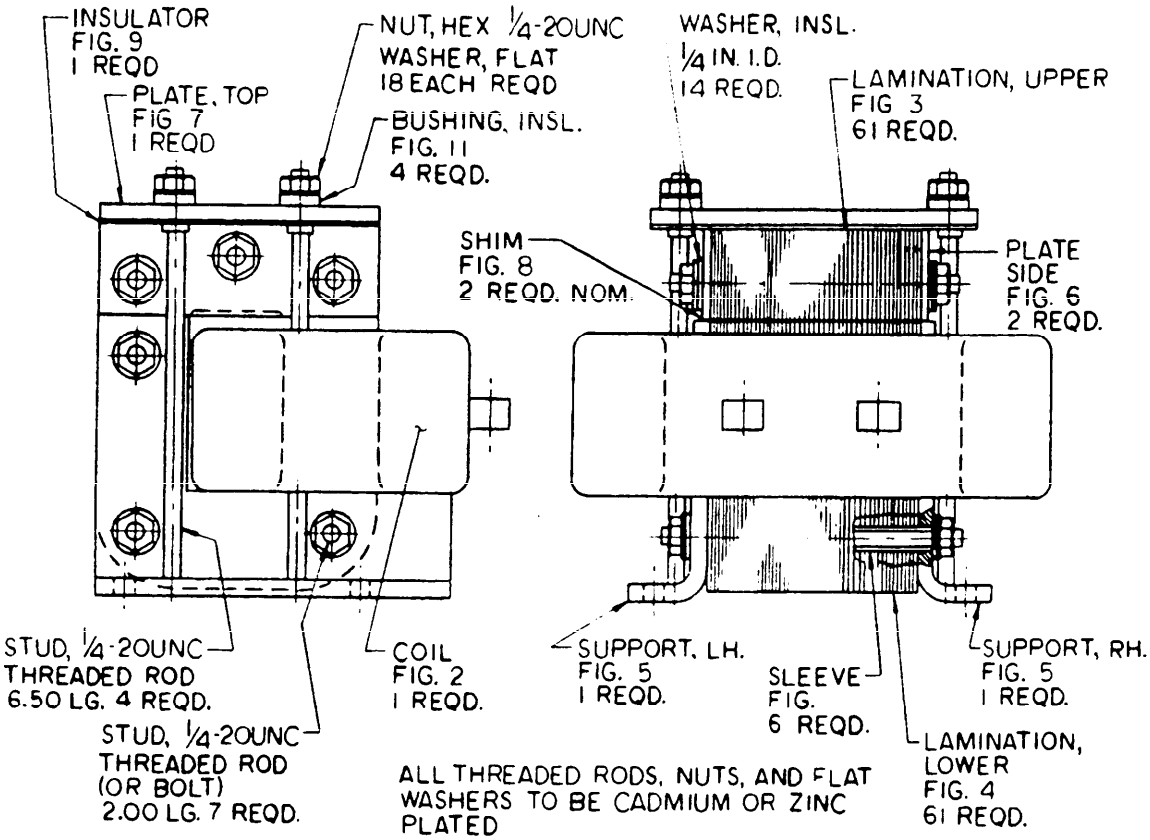
6.3 Definitions.

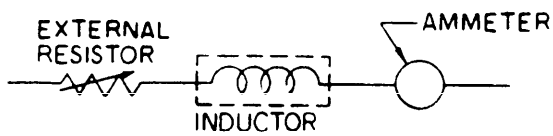
6.3.1 Response (current versus time). The current versus time ratio is a measure of the delayed rise of current due to self-inductance.

6.3.2 Stored energy. Stored energy is the energy in joules stored in the magnetic field, which must be dissipated when the switch under test opens the circuit.

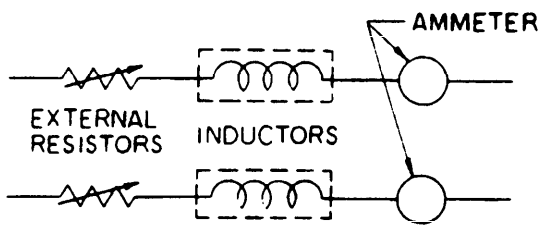
Preparing Activity:
Navy - AS
Project No. (6625-N835)

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CIRCUIT - SINGLE INDUCTOR



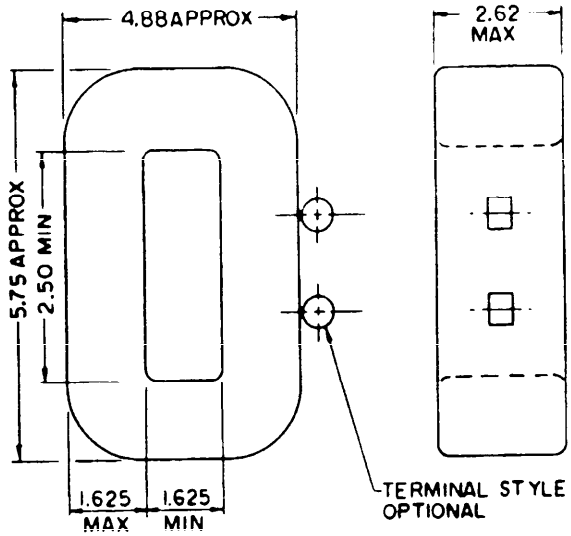
CIRCUIT - INDUCTORS IN PARALLEL (TYP)

NUMBER OF INDUCTORS IN PARALLEL	CURRENT RATING (AMPERES)
1	2.0-25.0
2	25.1-50.0
3	50.1-75.0
4	75.1-100.0

WHEN INDUCTORS ARE CONNECTED IN PARALLEL, CONNECT AS SHOWN AND ADJUST SERIES RESISTORS SO THAT THE CURRENTS THRU THE PARALLEL INDUCTORS ARE EQUAL WITHIN 3 PERCENT OF THE TOTAL CURRENT.

FIGURE 1. 28 volts D.C. inductor, type 1.

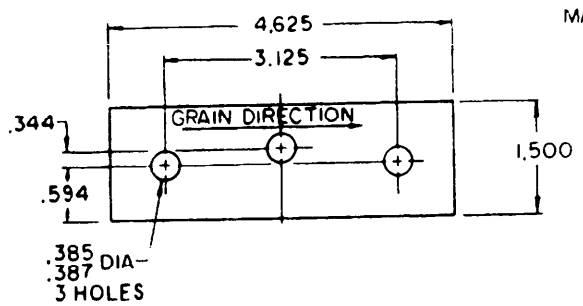
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COIL DATA	
LAYERS	13
TURNS LAYER	20
MEAN TURNS	13.4
SECTION SQ.IN.	3.8
VOLUME CU IN.	51
RESISTANCE OHMS	.286
WEIGHT LBS. (REF)	9.2
TURNS	255

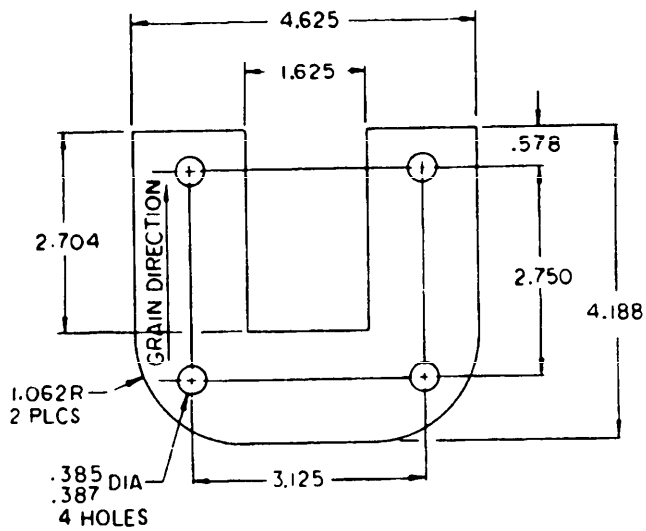
WIRE IS A.W.G. 10 PER FED SPEC J-W-1177/14, TYPE K2

FIGURE 2. Coil (1 required).



MATERIAL: .014 THICK ELECTRICAL STEEL SHEET PER AISI M-6 COAT WITH LAMINATION VARNISH

FIGURE 3. Lamination, upper (61 required).



MATERIAL: .014 THICK ELECTRICAL STEEL SHEET PER AISI M-6 COAT WITH LAMINATION VARNISH

FIGURE 4. Lamination, lower (61 required).

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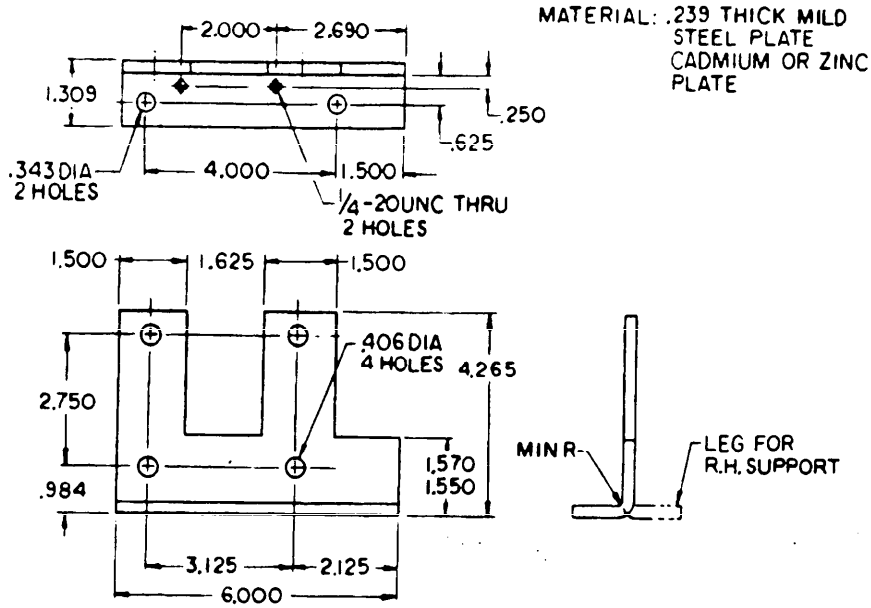


FIGURE 5. Support, LH/RH (1 each required).

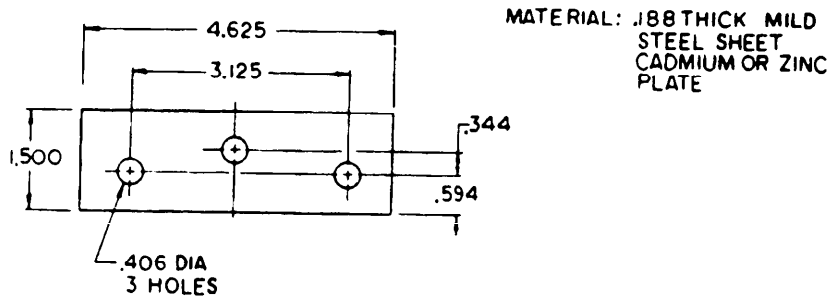


FIGURE 6. Plate, size (2 required).

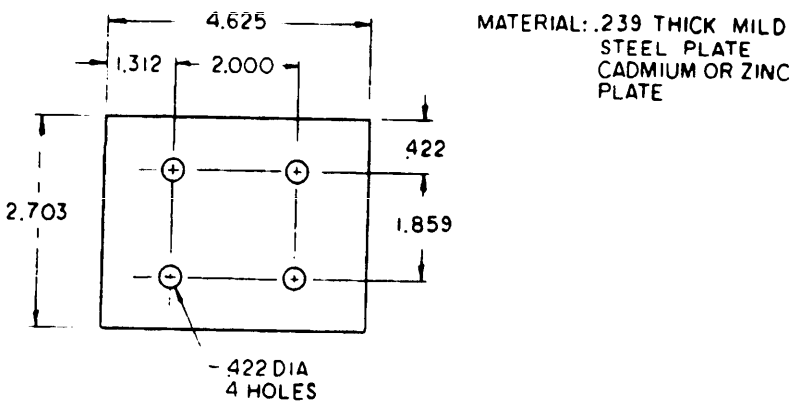


FIGURE 7. Plate, top (1 required).

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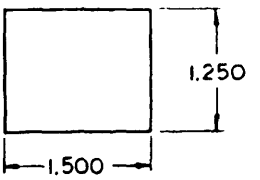


FIGURE 8. Shim (2 required nominally) (see paragraph).

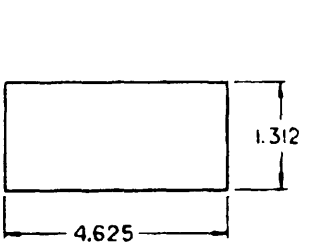


FIGURE 9. Insulator (1 required).

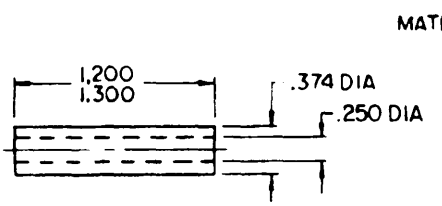


FIGURE 10. Sleeve (7 required).

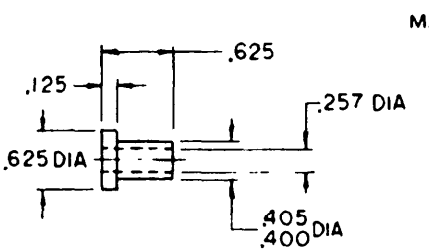
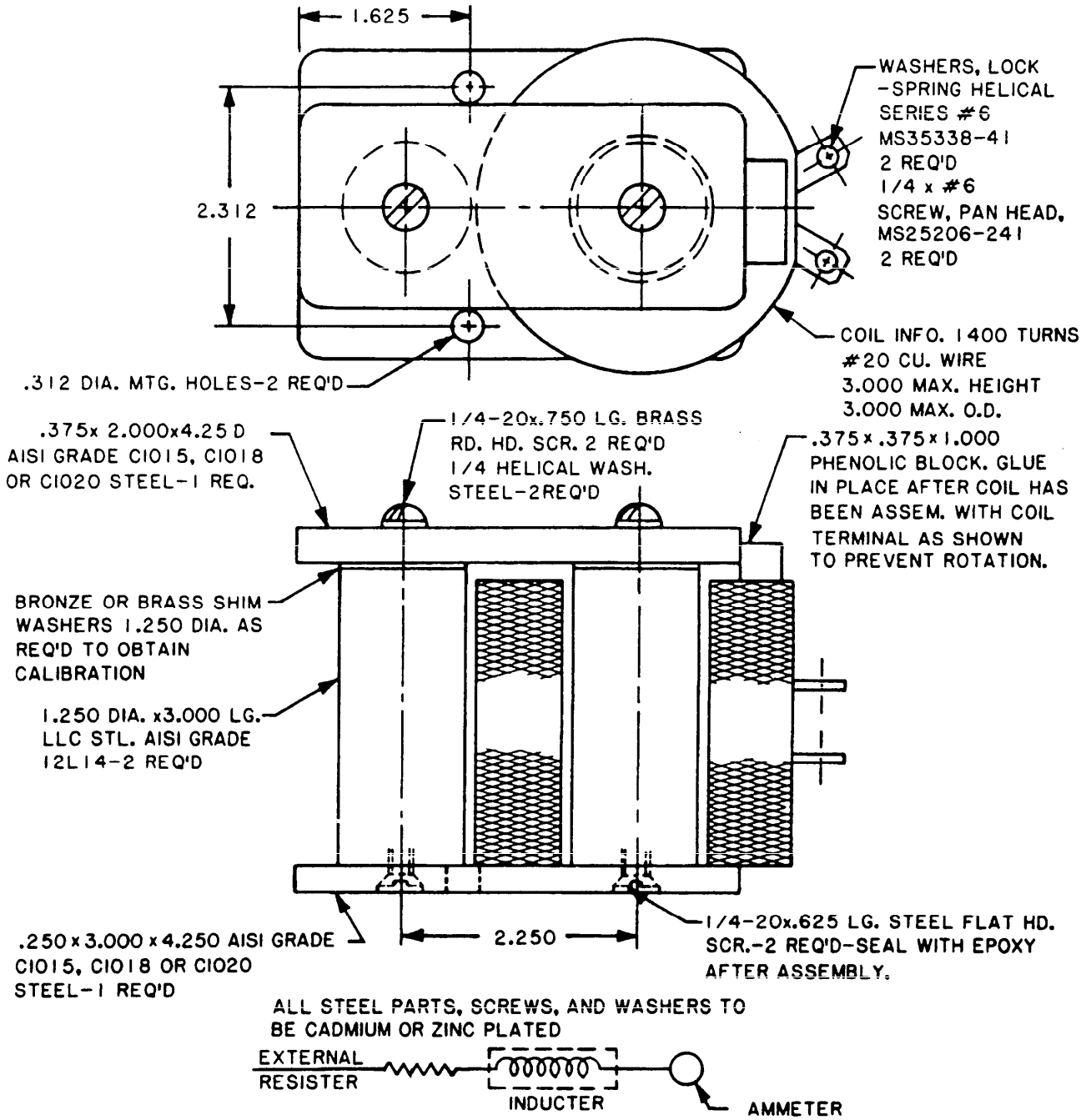


FIGURE 11. Bushing, insulating (4 required).

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PART NO. 81023-2

FIGURE 12. 28 volts D.C. inductor, type II.

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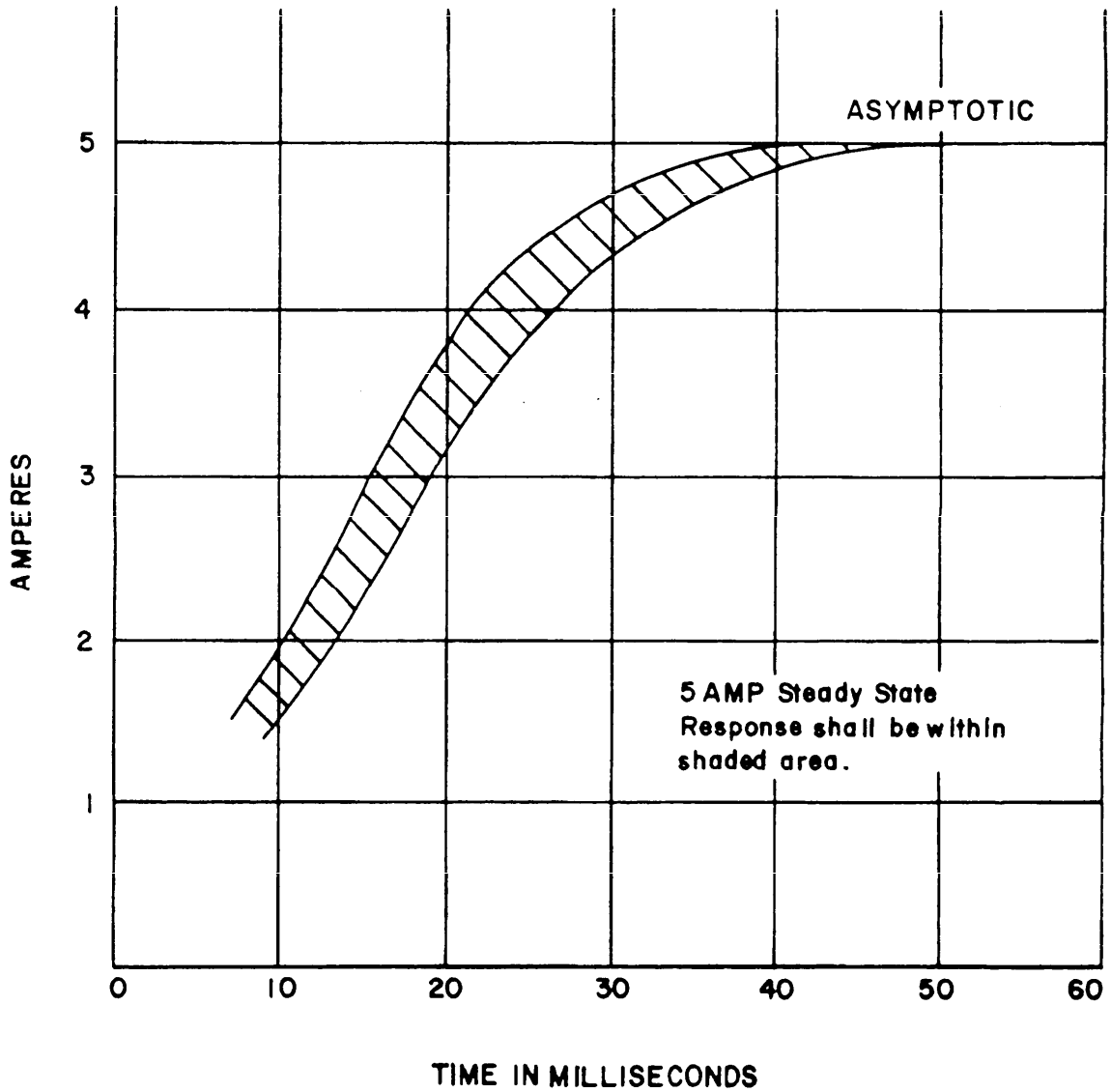


FIGURE 13. Response (current vs time) Type I inductors only.

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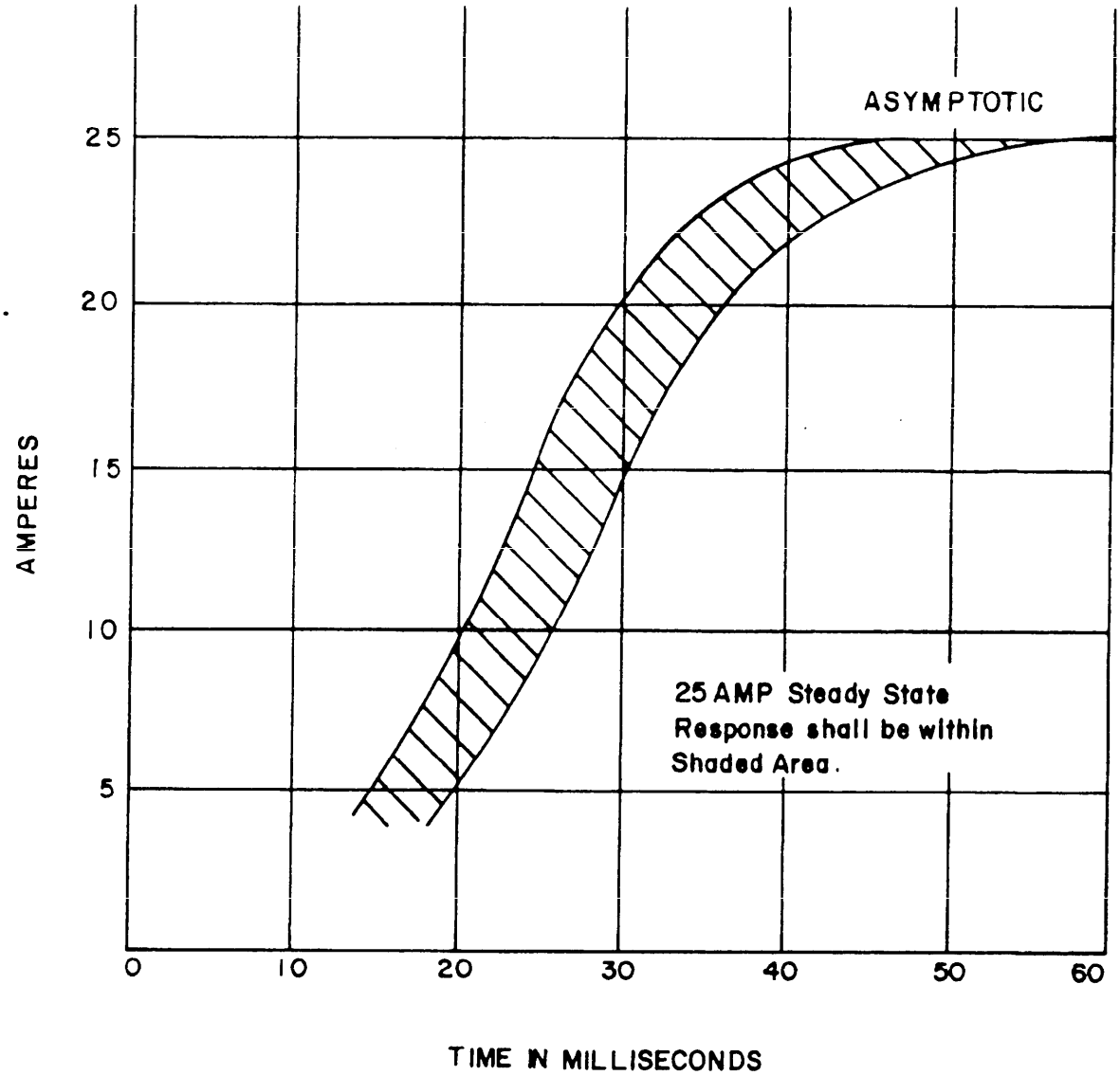


FIGURE 14. Response (current vs time) Type I inductors only.

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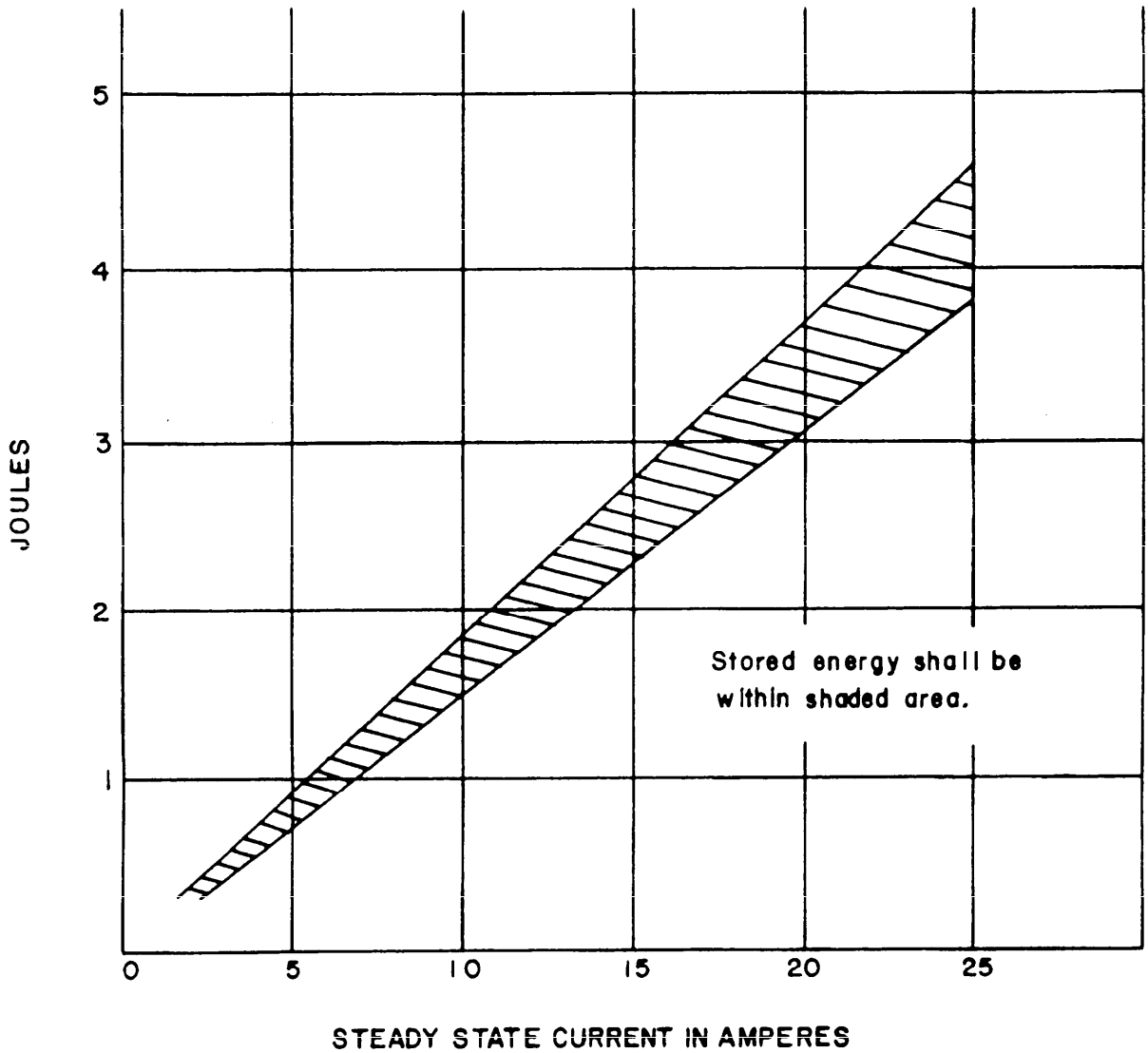


FIGURE 15. Stored energy (joules vs current) Type I inductors only.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-81023D	Inductor, 28 V.D.C. Laboratory Test General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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